

1 Prepare Labeled Dataset



Employee, Project and skill related data is merged and labeled.

2 Train / Test Split (80/20)



- Train : 1,800,000 records
- Test : 450,000 records

3 Balance Training Dataset



Imbalance handling (undersampling) to improve model learning.
Balanced Training Data:
18,864 records

4 Train HAF-EPA Model



Random Forest Classifier is trained on balanced data.

5 Save Trained Model



Model saved for future predictions.

HAF-EPA.joblib

6 Validate on 20% Test Dataset (Unseen Data)



Model is tested on unseen 450,000 records.

7 Evaluation Results



Model performance on test data.

Accuracy: 88.98%

8 Generate Hybrid Recommendations



Generate employee-project suitability recommendations on test dataset.

9 Create Output Files



- Feature importance (CSV)
- Test predictions (CSV)
- Recommendations (Excel)
- Evaluation report (TXT)
- Balanced training data (CSV)

10 Top 200 Employees Summary



Top 200 employees are selected based on overall compatibility and performance.

HAF-EPA_top_200_employees.xlsx

11 Employee Reference Database



Create top 100 employees ability and availability reference file.

HAF-EPA_employee_reference.csv

12 Used by Web Application



These outputs are used by the web application to recommend the best suitable employees for new projects.

1. Prepare Labeled Dataset

Merge employee, project, and skill data. Create labeled records for model training.

2. Train/Test Split (80/20)

Split dataset into training (80%) and testing (20%). Train: 1.8M records, Test: 450K records.

3. Balance Training Dataset

Apply undersampling to handle class imbalance. Balanced training set: 18,864 records.

4. Train HAF-EPA Model

Train a Random Forest classifier. Use balanced data for learning patterns.

5. Save Trained Model

Store the trained model using Joblib. Output: HAF-EPA.joblib.

6. Validate on Test Dataset

Evaluate the model on unseen test data. Tested using 450,000 records.

7. Evaluation Results

Measure model performance on test data. Achieved accuracy of 88.98%.

8. Generate Hybrid Recommendations

Generate employee-project matching recommendations. Rank employees based on suitability scores.

9. Create Output Files

Export predictions, reports, and recommendations. Save results in CSV, XLSX, and TXT formats.

10. Top 200 Employees Summary

Select top employees based on compatibility. Export summary to Excel file.

11. Employee Reference Database

Create employee reference and availability data. Store key employee information for deployment.

12. Used by Web Application

Web application consumes generated outputs. Provides employee recommendations for projects.